

Genetic report for longevity

Spine, Disc and Low Back Pain

Degeneration and herniation of the disc, lumbar and sciatic pain, stenosis, scoliosis, ankylosing spondylitis, chronic pain

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The intervertebral disc is the tissue that ages earlier: it loses water and proteoglycans (aggrecan) from the 20–30 years, and the degeneration of the disk is about 30–75% hereditary. This report reads the disk collagens (COL9A2/COL9A3 "Trp", COL11A1), CHST3, CILP and GDF5, lumbar and sciatic pain loci (SOX5, DCC, CCDC26/GSDMC), the marker of HLA-B27 (ankylosing spondylitis), scoliosis, stenosis and chronic pain in several locations. The disc hernia score (the 3rd percentile) is already in the Extracellular Matrix and Glycation report.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

23

variants

15

in the chip

6

imputed
(DR2 $\geq$ 0.8)

2

not found

9

Scores with
percentile

Overall reading. The column has clearly favourable genetics. In the disc: score of intervertebral disc disease at the 17th percentile, disc herniation in the 40 (PGS) and in the 3 (Loci score of the Extracellular Matrix report), disc degeneration in image in the 15 (only 3 loci). Without the Trp2 alleles of COL9A2 and Trp3 of COL9A3, CHST3 without risk alleles in both signals, GDF5 C/C in the gene (without the T allele that lowers its expression) and SPOCK2 in favor; the only point against is 1 allele of COL11A1, with weak evidence. In pain: the score of chronic low back pain is at percentile 0.6 (39 loci, allele count), DCC without risk alleles and GSDMC with 1 (typical); chronic pain in several places is in the middle (51). In the structure, stenosis in 28 and spondylosis in 56; LBX1 TT pulls scoliosis upwards, but it is a disease of adolescence and the score of scoliosis is in 55. In inflammatory pain: probable negative HLA-B27 and ERAP1 without the 528R allele; the score of ankylosing spondylitis is in 73, but without B27 the absolute risk is very low. The one that counts the most in the column from here on is mechanical and controllable: trunk and hip strength, training technique, sitting time and keeping lean mass during retatrutide.

Favourable

- **Intervertebral disc disease (111 loci):** the 17th percentile
- **Chronic low back pain:** the 1st percentile
- **Low back disc hernia (loci score):** the 3rd percentile (Extracellular Matrix and Glycation)
- **COL9A2 (Trp2) rs12077871 GG:** COL9A2 Gln326Trp: Trp2 allele, associated with degeneration and disc herniation, especially in the initial study.
- **CHST3 rs4148941 CC:** CHST3: aggrecan sulfate; A allele lowers expression and is associated with disc degeneration (OR $\sim$ 1.3).
- **GDF5 rs143383 GG:** GDF5 +104T/C: T allele lowers the expression of GDF5; osteoarthritis and disc degeneration.
- **DCC rs4384683 AA:** DCC: Nethrin receptor, axon guide; low back pain and chronic pain.
- **HLA-B27 (tag) rs4349859 GG:** HLA-B27 (allele A); about 90% of patients with ankylosing spondylitis are B27 positive.
- **ERAP1 rs30187 CC:** ERAP1 K528R: T allele (528R) increases the risk of spondylitis in positive B27.
- **CHST3 rs4284332 TT:** CHST3: main variant in GWAS.
- **SPOCK2 rs11000138 TT:** SPOCK2: proteoglycan; stenosis and hernia.

Unfavourable

- **LBX1 rs11190870 TT:** LBX1: the strongest locus of idiopathic scoliosis.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 15 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Intervertebral disc: degeneration and hernia

The core of the disc is aggrecan and collagen II; the ring is collagen I. The Trp alleles of COL9A2 and COL9A3, COL11A1, CHST3 (sulfation of aggrecan) and CILP move the resistance of the disc. The VNTR polymorphism of aggrecan (ACAN) is not read by chip.

Ageing mechanisms: Remodelling of the extracellular matrix; Stiffness of the extracellular matrix; Accumulation of Advanced Glycation Final Products (AGEs)

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Intervertebral disc disease (111 loci)	Favourable	the 17th percentile · z=-0.95	Higher score : higher risk. PGS000932 (PGS Catalog; penalized regression; Other intervertebral disk disorders (time-to-event); PGP000244, PMID 35324888). Variants with genotype: 93 of 111 (covering of 90.9% coefficients).
Disc hernia (47 loci)	Typical / intermediate	the 40th percentile · z=-0.31	Higher score : higher risk. PGS000964 (PGS Catalog; penalized regression; Slipped disc; PGP000244, PMID 35324888). Variants with genotype: 39 of 47 (covering of 90.6% coefficients).
Discal degeneration (image)	Typical / intermediate	the 32nd percentile · z=-0.48	Higher score : higher risk. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 3 of 3 loci with genotype. Studies: Williams FM (Disc degeneration) Variants with genotype: 3 of 3 (covering of 100.0% coefficients).
Low back disc hernia (loci score)	Favourable	the 3rd percentile	GWAS Catalog loci score; few loci. Source: Extracellular Matrix and Glycation report.

Favourable

- Intervertebral disc disease (111 loci) (the 17th percentile)
- Low back disc hernia (loci score) (the 3rd percentile)
- COL9A2 (Trp2) rs12077871 GG
- COL9A3 (Trp3) rs61734651 CC
- CHST3 rs4148941 CC
- GDF5 rs143383 GG

Unfavourable

- COL11A1 rs1676486 AG

Favourable variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL9A2 (Trp2)	rs12077871 1:40307478 · G/A/T	GG	chip	A (0.0; 0%)	moderate	COL9A2 Gln326Trp: Trp2 allele, associated with degeneration and disc herniation, especially in the initial study. Without the Trp2 allele (A in the strand read; the gene is in reverse strand).
COL9A3 (Trp3)	rs61734651 20:62819980 · C/T	CC	chip	T (0.0; 6%)	weak	COL9A3 Arg103Trp: allele Trp3; partially replicated association. Without the Trp3 allele.
CHST3	rs4148941 10:72009234 · C/A/T	CC	imputation DR2=1.00	A (0.0; 57%)	moderate	CHST3: aggrecan sulfate; A allele lowers expression and is associated with disc degeneration (OR ~1.3).
GDF5	rs143383 20:35438203 · G/A	GG	chip	A (0.0; 63%)	moderate	GDF5 +104T/C: T allele lowers the expression of GDF5; osteoarthritis and disc degeneration. GG = C/C in the gene: without the risk T allele.

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
COL11A1	rs1676486 1:102888582 · A/G/T	AG	chip	A (1.0; 17%)	weak	COL11A1 c.4603C>T: T allele lowers the expression of COL11A1 in the disc; lumbar hernia in studies of a single population. Gene in reverse strand: the T allele of the gene is A in the strand read.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PARK2	rs926849 6:161740587 · C/T	CT	chip	T (1.0; 70%)	weak	PARK2: disc degeneration locus in an image GWAS (MRI). Meaning: GWAS Catalog: Disc degeneration (lumbar) (Williams FM 2012; p=3e-08); C ↓ allele

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CILP	rs2073711 15:65201874 · A/G	GG	chip	G (2.0; 65%)	weak	CILP 1184T>C: inhibits TGF-β in the disc; lumbar hernia in a study, without consistent replication. Gene in reverse strand: the C allele of the gene is G in the strand read. GG = 2 copies of the allele described as risk, which is the most common in the reference population; association not replicated, therefore without judgment.

Low back pain, sciatic pain and chronic pain

The GWAS of chronic low back pain find SOX5, CCDC26/GSDMC and DCC (axon guide); sciatic sharing loci with hernia. Chronic pain in several places is very polygenic and linked to mood and sleep.

Ageing mechanisms: Neuroinflammation; Chronic Inflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Chronic low back pain	Favourable	the 1st percentile · z=-2.20	Higher score : higher risk. Loci score of GWAS Catalog ($p\leq 510-8$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 33 of 39 loci with genotype. Studies: Bjornsdottir G (ICD10 M54: Dorsalgia); Freidin MB (Back pain); Sakaue S (Back pain); Suri P (Chronic back pain); Verma A (Back pain (PheCode 760)). Variants with genotype: 33 of 39 (covering of 84.6% coefficients).
Chronic pain in several locations	Typical / intermediate	the 51st percentile · z=0.06	Higher score : higher risk. GWAS Catalog loci score ($p\leq 10-6$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 25 of 25 loci with genotype. Studies: Johnston KJA (Multisite chronic pain); Verma A (Chronic pain (PheCode 338.2)). Variants with genotype: 25 of 25 (covering of 100.0% coefficients).

Favourable

- Chronic low back pain (the 1st percentile)
- DCC rs4384683 AA
- CHST3 rs4284332 TT
- SPOCK2 rs11000138 TT

Unfavourable

None.

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DCC	rs4384683 18:52852662 · G/A/C/T	AA	imputation DR2=0.99	G (0.0; 45%)	moderate	DCC: Nethrin receptor, axon guide; low back pain and chronic pain. Meaning: GWAS Catalog: Chronic back pain (Suri P 2018; $p=2e-10$); G ↑ allele
CHST3	rs4284332 10:71974194 · T/A/C	TT	imputation DR2=0.99	C (0.0; 58%)	moderate	CHST3: main variant in GWAS. Meaning: GWAS Catalog: Lumbar disc herniation (Salo V 2024; $p=1e-31$); allele C ↑
SPOCK2	rs11000138 10:72022264 · A/C/T	TT	imputation DR2=0.99	A (0.0; 57%)	moderate	SPOCK2: proteoglycan; stenosis and hernia. Meaning: GWAS Catalog: Displacement of intervertebral disc (PheCode 722.1) (Verma A 2024; $p=3e-13$); Allele A ↑

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CCDC26/ GSDMC	rs7833174 8:129706526 · T/C/G	CT	imputation DR2=1.00	T (1.0; 74%)	moderate	GSDMC: the most replicated locus of lumbar pain and disc herniation. Meaning: GWAS Catalog: Chronic back pain (Suri P 2018; $p=4e-13$); T ↑ allele
SCN9A	rs6746030 2:166242648 · A/C/G	GG	chip	A (0.0; 13%)	weak	SCN9A R1150W: Allele A increases the excitability of Nav1.7 and pain sensitivity.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Evidence	Interpretation
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				Unfavourable allele (copying; frequency in reference)		
SOX5	rs34717197 7:152029908 · C/T	CC	imputation DR2=0.88	—	moderate	SOX5: chondrogenesis; locus of chronic low back pain.
OPRM1	rs1799971 6:154039662 · A/G	AG	chip	—	weak	OPRM1 A118G: the G allele asks for more opioids for the same effect. Information.
COMT	rs4680 22:19963748 · G/A	GG	chip	—	weak	COMT Val158Met Met/Met was associated with more sensitivity to pain in small studies. Information.
VDR	rs731236 12:47844974 · A/G/T	AG	chip	—	weak	VDR TaqI: old and inconsistent associations with dyschalc degeneration. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FAM20C	— — · —	—	SNP not found in the data	—	moderate	FAM20C. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Structure: stenosis, scoliosis and spondylosis

Low back stenosis and spondylosis are degeneration of the disc and facets with age; idiopathic scoliosis has its own loci (LBX1, GPR126/ADGRG6). The vertebral bone density is in the Musculus, Bone and Sarcopenia report.

Ageing mechanisms: Osteoporosis; Fragility

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Lumbar stenosis	Typical / intermediate	the 28th percentile · z=-0.68	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 8 of 8 loci with genotype. Studies: Verma A (Spinal stenosis (PheCode 720)); Verma A (Spinal stenosis of lumen region (PheCode 720.1)). Variants with genotype: 8 of 8 (covering of 100.0% coefficients).
Scoliosis	Typical / intermediate	the 55th percentile · z=0.07	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 24 of 30 loci with genotype. Studies: Kou I (Adolescent idiopathic scoliosis); Miyake A (Adolescent idiopathic scoliosis (severe)); Ogura Y (Curve progression in adolescent idiopathic scoliosis); Verma A (Kyphoscoliosis and scoliosis (PheCode 737.3); Zhu Z (Adolescent idiopathic scoliosis). Variants with genotype: 24 of 30 (covering of 80.0% coefficients).
Spondylosis (cervical and lumbar)	Typical / intermediate	the 56th percentile · z=0.48	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 4 of 5 loci with genotype. Studies: Taniguchi Y (Severe lumen spondylosis); Verma A (Spondylosis and allied disorders (PheCode 721)); Walters RG (ICD10 M47: Spondylosis); Zhang Y (Spondylosis). Variants with genotype: 4 of 5 (covering of 80.0% coefficients).

Favourable

None.

Unfavourable

- LBX1 rs11190870 TT

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LBX1	rs11190870 10:101219450 · T/A/C	TT	chip	T (2.0; 59%)	strong	LBX1: the strongest locus of idiopathic scoliosis. Meaning: GWAS Catalog: Adolescent idiopathic scoliosis (Kou I 2019; p=2e-82); T ↑ allele

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADGRG6	rs6570507 6:142358435 · G/A	GG	chip	—	moderate	ADGRG6 (GPR126): scoliosis and growth.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SLC26A2	— · —	—	SNP not found in the data	—	moderate	SLC26A2: cartilage sulfate transporter. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Ankylosing spondylitis and inflammatory pain

Ankylosing spondylitis is an inflammatory low back pain (worst at rest and in the morning). About 90% of patients have HLA-B27; ERAP1 and IL23R modulate the risk.

Ageing mechanisms: Chronic Inflammation; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Ankylosing spondylitis (85 variants)	Typical / intermediate	the 73rd percentile · z=0.21	Higher score : higher risk. PGS001876 (PGS Catalog; Penalized regression; Ankylosing spondylitis; PGP000263, PMID 34995502). Variants with genotype: 81 of 85 (covering of 75.0% coefficients).

Favourable

- HLA-B27 (tag) rs4349859 GG
- ERAP1 rs30187 CC

Unfavourable

None.

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HLA-B27 (tag)	rs4349859 · 6:31398010 · G/A/C	GG	chip	A (0.0; 5%)	strong	HLA-B27 (allele A); about 90% of patients with ankylosing spondylitis are B27 positive. Likely negative HLA-B27 (tag; is not HLA typing).
ERAP1	rs30187 · 5:96788627 · T/A/C	CC	chip	T (0.0; 35%)	moderate	ERAP1 K528R: T allele (528R) increases the risk of spondylitis in positive B27.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL23R	rs11209026 · 1:67240275 · G/A	GG	chip	G (2.0; 94%)	strong	IL23R R381Q: allele A (381Q) protects; G is the common.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Accumulation of Advanced Glycation Final Products (AGEs)	COL11A1 AG	Intervertebral disc disease (111 loci) (the 17th percentile); Low back disc hernia (loci score) (the 3rd percentile); COL9A2 (Trp2) GG; COL9A3 (Trp3) CC; CHST3 CC; GDF5 GG	Disc hernia (47 loci) (the 40th percentile); Discal degeneration (image) (the 32nd percentile)
Fragility	LBX1 TT	—	Lumbar stenosis (the 28th percentile); Scoliosis (the 55th percentile); Spondylosis (cervical and lumbar) (the 56th percentile)
Osteoporosis	LBX1 TT	—	Lumbar stenosis (the 28th percentile); Scoliosis (the 55th percentile); Spondylosis (cervical and lumbar) (the 56th percentile)
Remodelling of the extracellular matrix	COL11A1 AG	Intervertebral disc disease (111 loci) (the 17th percentile); Low back disc hernia (loci score) (the 3rd percentile); COL9A2 (Trp2) GG; COL9A3 (Trp3) CC; CHST3 CC; GDF5 GG	Disc hernia (47 loci) (the 40th percentile); Discal degeneration (image) (the 32nd percentile)
Stiffness of the Extracellular Matrix	COL11A1 AG	Intervertebral disc disease (111 loci) (the 17th percentile); Low back disc hernia (loci score) (the 3rd percentile); COL9A2 (Trp2) GG; COL9A3 (Trp3) CC; CHST3 CC; GDF5 GG	Disc hernia (47 loci) (the 40th percentile); Discal degeneration (image) (the 32nd percentile)
Intercellular Communication	—	HLA-B27 (tag) GG; ERAP1 CC	Ankylosing spondylitis (85 variants) (the 73rd percentile); IL23R GG
Chronic Inflammation	—	Chronic low back pain (the 1st percentile); DCC AA; CHST3 TT; SPOCK2 TT; HLA-B27 (tag) GG; ERAP1 CC	Chronic pain in several locations (the 51st percentile); CCDC26/GSDMC CT; Ankylosing spondylitis (85 variants) (the 73rd percentile); IL23R GG
Neuroinflammation	—	Chronic low back pain (the 1st percentile); DCC AA; CHST3 TT; SPOCK2 TT	Chronic pain in several locations (the 51st percentile); CCDC26/GSDMC CT

Practical part

Ideas for the spine, crossed with the genetic plan v2 and current medication (isotretinoin, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin). The current stack remains as it is; everything is suggestion (not applied). With this genetics, the objective is to maintain: the disks feed by diffusion and respond well to varied and regular load.

1. Walking every day and interrupting sitting time

Genetics: Low back pain p0.6; p17 disk

Walking or running regularly is associated with more hydrated discs on MRI. Stand and walk 2–3 minutes every 30–45 minutes sitting; 7–10 thousand steps per day.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

2. Force of the trunk and hips, with technique

Genetics: CHST3, COL9 and GDF5 in favor; COL11A1 with 1 allele

Maintain strength training 3×/week with hip fold exercises (Romanian land lift, hip thrust), board and anti-rotation variants. In the bending of the loaded column, prioritize the technique over the load.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

3. Retatrutide: protecting lean mass

Genetics: Low genetic lean mass (Muscle, Bone and Sarcopenia)

A large caloric deficit also removes muscle to the stabilizers of the spine. Protein ~ 140 g/day, creatine (already taken) and strength training throughout the treatment.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

4. NSAID: low and short dose

Genetics: CYP2C9 *1/*3 (pharmacological profile)

CYP2C9 *1/*3 slowly eliminates ibuprofen, celecoxib and especially meloxicam (CPIC guide: starting with the lowest dose; meloxicam with half of the dose). With telmisartan and dapagliflozin, regular NSAID overload the kidney: use only in short and well hydrated seizures.

V2 genetic plan and medication: No conflict; reinforces the caution of the pharmacogenetic profile. • suggestion (not applied)

5. Isotretinoin: attention to new low back pain

Genetics: probable negative HLA-B27

Isotretinoin gives muscle and joint pain and rarely sacroiliitis. New lumbar pain that worsens at rest and improves with movement, with morning stiffness above 30 minutes, requires medical evaluation even without B27.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

6. If one day you need strong analgesia

Genetics: OPRM1 A/G; COMT Val/Val

The OPRM1 A/G can request a little more opioid for the same effect; COMT Val/Val is associated with less pain sensitivity. It is informative for surgery, it does not change anything today.

V2 genetic plan and medication: No conflict. • suggestion (not applied)

Avoiding or with caution

- NSAID daily for weeks (CYP2C9 *1/*3; kidney with telmisartan and dapagliflozin). suggestion (not applied)
- Sitting for hours without pauses. suggestion (not applied)
- Training with maximum bending load of the column without technique. suggestion (not applied)
- Long-term bed rest in a low back pain crisis: staying active is the one that accelerates recovery. suggestion (not applied)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. The VNTR of aggrecan and 5A/6A of MMP3 are not read; the HLA-B27 is inferred by a tag.
- Many candidate genes of the disc (COL9, COL11A1, CILP, VDR) came from small studies with weak replication; the loci of GWAS (GSDMC, SOX5, DCC, CHST3) are more solid, but with small effect.
- Low back pain depends heavily on mechanical load, training, sleep and mood; the genes explain a small part.
- The sciatic score was not interpreted (2 loci, weighted coverage of 4%).
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS000932 (PGS Catalog; penalized regression; Other intervertebral disk disorders (time-to-event); PGP000244, PMID 35324888 · PGS000964 (PGS Catalog; penalized regression; Slipped disc; PGP000244, PMID 35324888 · PGS001876 (PGS Catalog; Penalized regression; Ankylosing spondylitis; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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Generated in 02-10-2026 (local time) by the scripts of this report.